A FIBRE PRODUCT OF ALGEBRAIC SPACES OVER A SHEAF WITH REPRESENTABLE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains one xymatrix display; it is part of the counted proof span and is preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Definition of an algebraic space. Definition definition-algebraic-space, used at proof lines 1140-1141 and 1168-1170. Supplies the three conditions that are verified.
- Products of representable transformations are representable. Lemma lemma-product-representable-transformations (409-420), used at proof lines 1152-1155. Makes Delta_F x Delta_G representable.
- Base change of a representable transformation is representable. Lemma lemma-base-change-representable-transformations, used at proof lines 1152-1155 and 1163-1167. Carries representability to the diagonal and to the chart.
- Composition of representable transformations with a property. Lemma lemma-composition-representable-transformations, used at proof lines 1163-1167. Gives the surjective etale chart.

Source reference. Original definition 025Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2831–2831. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3798–3798. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3797–3797. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02WC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3799–3799. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let H be a sheaf on $(Sch/S)_{fppf}$ whose diagonal is representable. Let F, G be algebraic spaces over S. Let $F \to H$, $G \to H$ be maps of sheaves. Then $F \times_H G$ is an algebraic space.*

Proof. We check the 3 conditions of Definition 025Y. A fibre product of sheaves is a sheaf, hence $F \times_H G$ is a sheaf. The diagonal of $F \times_H G$ is the left vertical arrow in

$$\begin{array}{ccc} F \times_H G & \longrightarrow & F \times G \\ {\scriptstyle \Delta}\downarrow & & \downarrow{\scriptstyle \Delta_F \times \Delta_G} \\ (F \times F) \times_{(H \times H)} (G \times G) & \longrightarrow & (F \times F) \times (G \times G) \end{array}$$

which is cartesian. Hence Δ is representable as the base change of the morphism on the right which is representable, see Lemmas 02WC and 02WB. Finally, let $U, V \in \mathrm{Ob}((Sch/S)_{fppf})$ and $a : U \to F$, $b : V \to G$ be surjective and étale. As Δ_H is representable, we see that $U \times_H V$ is a scheme. The morphism

$$U \times_H V \longrightarrow F \times_H G$$

is surjective and étale as a composition of the base changes $U \times_H V \to U \times_H G$ and $U \times_H G \to F \times_H G$ of the étale surjective morphisms $U \to F$ and $V \to G$, see Lemmas 02WA and 02WB. This proves the last condition of Definition 025Y holds and we conclude that $F \times_H G$ is an algebraic space. □